Handling Strings

Accessing Values in Strings

Python does not support a character type; these are treated as strings of length one, thus also considered a substring.
To access substrings, use the square brackets for slicing along with the index or indices to obtain your substring. For example −

```
var1 = 'Hello World!'
var2 = "Python Programming"
print "var1[0]: ", var1[0]
print "var2[1:5]: ", var2[1:5]
```

String Special Operators

Assume string variable **a** holds 'Hello' and variable **b** holds 'Python', then −

Operator	Description
+	Concatenation - Adds values on either side of the operator
*	Repetition - Creates new strings, concatenating multiple copies of the same string
[]	Slice - Gives the character from the given index
[:]	Range Slice - Gives the characters from the given range
In	Membership - Returns true if a character exists in the given string
not in	Membership - Returns true if a character does not exist in the given string
%	Format - Performs String formatting

Problem Statement:

1. Write a program which will print each character of a string.

```
sampleStr = "Hello!!"
print("**** Iterate over string using for loop****")
for elem in sampleStr:
    print(elem)
```

2. Write a program to count a particular letter in a string.

```
sampleStr = "Hello!!"
count=0
print("**** Iterate over string using for loop****")
for elem in sampleStr:
        if(elem==’l’)
     count=count+1
print(count)
```

b) Prints the last three characters of a string.

```
>>> st=”Python”
>>> st[-3:]
'hon'
```

c) Write a string, replace its first letter.

```
>>> st="Python"
>>> 'J'+st[1:]
'Jython'
```

d) Write a string include two more letters after the first two characters.

```
>>> st="Python"
>>> st[0:2]+st[-2:]
'Pyon'
```

e) Save a sentence in a variable. Print every third position character from first.

```
>>>st=”he ate camel food”
>>>st[::3]
‘ha m o’
```

f) Save a sentence in a variable. Print every second position character from last.

```
>>>st=”he ate camel food”
>>>st[::-2]
‘do ea t h’
```

g) Save a sentence in a variable. Print it in reverse without using the function.

```
>>> s='he ate camel food'
>>> s[::-1]
'doof lemac eta eh'
```

h) Store a list of strings. Print it by joining all of them using “-<>-”.

```
>>> list=['hello','welcome','bye']
>>> "-<>-".join(list)
'hello-<>-welcome-<>-bye'
```

i) Store a character in a variable in such a way that it will be printed five times. (Do it in a single statement)

```
>>> s="="*5
>>> print(s)
=====
```

j) Mention width and print a string with center aligned.

```
>>> s='hello'
>>> s.center(10,' ')
'  hello   '
```

k) Count the number of occurrences of a string in another string.

```
>>> s='hello'
>>> s.count('l')
2
```

l) Print leftmost position of matched substring. What will be the output if the substring is not matched Check the result.

```
>>> s="Welcome to the world of wonder"
>>> x1="ld"
>>> s.find(x1,0,len(s))
18
>>> x="ly"
>>> s.find(x,0,len(s))
-1
```

m) Mention a string and substring which is present in the string. Now partition it.

```
>>> s="Welcome to the world of Python Jython"
>>> t="thon"
>>> s.partition(t)
('Welcome to the world of Py', 'thon', ' Jython')
>>> s='madam'
>>> t='ma'
>>> s.partition(t)
('', 'ma', 'dam')
```

n) Replace a substring by another substring in a string.

```
>>> s="Welcome to the world of Python Jython"
>>> t='on'
>>> u='xxyy'
>>> s.replace(t,u)
'Welcome to the world of Pythxxyy Jythxxyy'
>>> t1='x'
>>> s.replace(t1,u)
'Welcome to the world of Python Jython'
```

o) Split a sentence w.r.t. a substring mentioning both one and two parameters.

```
>>> s="Welcome to the world of Python Jython"
>>> t="thon"
>>> s.split(t)
['Welcome to the world of Pyth', ' Jyth', '']
>>> s.split(t,1)
['Welcome to the world of Pyth', ' Jython']
```

p) Do the above for thrice from start.

```
>>> s="Welcome to the world of Python Jython"
>>> x='o'
>>> u=' '
>>> s.split(x,3)
['Welc', 'me t', ' the w', 'rld of Python Jython']
```

q) Do the above for thrice from end.

```
>>> s="Welcome to the world of Python Jython"
>>> x='o'
>>> u=' '
>>> s.rsplit(x,3)
['Welcome to the world ', 'f Pyth', 'n Jyth', 'n']
```

r) Write a sentence having a name, date of birth, date of death separated by * without space. Split it and store it in a list. Split the dates and save it in two separate lists. Print as follows:

XYZ lived about 82 years

(Name and age are variables) (Dates are written as YYYY-MM-DD)

```
>>> record='Leo Tolstroy*1828-8-28*1910-11-20'
>>> fields=record.split("*")
>>> fields
['Leo Tolstroy', '1828-8-28', '1910-11-20']
>>> born=fields[1].split("-")
>>> born
['1828', '8', '28']
>>> died=fields[2].split("-")
>>> died
['1910', '11', '20']
>>> print("lived about",int(died[0])-int(born[0]),"years")
lived about 82 years
```

u) Using “in” operator check whether a substring is present in the string.

```
>>> 'a' in 'banana'
True
>>> t='a'
>>> s='banana'
>>> t in s
True
```

v) Capitalize the first letter.

```
>>> s='hello'
>>> s.capitalize()
'Hello'
```

w) Check whether a tuple of substring are present in a string and substring slice. Provide at least five different instances.

```
>>> s="Welcome to the world of wonder"
>>> x=("me","ld","er")
>>> s.endswith(x,0,30)
True
>>> s.endswith(x,0,len(s))
True
>>> s.endswith(x,0,15)
False
>>> x1="ld"
>>> s.endswith(x1,0,len(s))
False
>>> x=("me","ld")
>>> s.endswith(x,0,len(s))
False
```

x) Split lines which are in triple quote. Also, split another lines which includes escape sequences.

```
>>> s='''welcome dear
how are you?
Where do you live?
Do you love Kolkata?'''
>>> s.splitlines()
['welcome dear', 'how are you?', 'Where do you live?', 'Do you love Kolkata?']
>>> g='welcome\n dear\r milk \t chocolate'
>>> g.splitlines()
['welcome', ' dear', ' milk \t chocolate']
>>> g.splitlines(True)
['welcome\n', ' dear\r', ' milk \t chocolate']
```

y) Check the starting substring of a string for different instances.

```
>>> s="Welcome to the world of Python Jython"
>>> t=('a',"W",'t')
>>> s.startswith(t)
True
>>> t1='w'
>>> s.startswith(t1)
False
```